

Adaptive Process Noise Estimation in Sim2Real Frameworks: A Comparative Analysis Across Nonlinear Kalman Filters

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Abstract—Accurate underwater navigation is challenging due to the unavailability of GNSS signals and time-varying uncertainties in inertial and Doppler velocity log (DVL) measurements. This paper proposes SINNE-KF, a hybrid adaptive navigation framework that combines innovation-based covariance adaptation with neural estimation of time-varying inertial sensor noise. The method is integrated into three filtering frameworks: SINNE-EKF, SINNE-UKF, and SINNE-RI-IKF. The approaches are evaluated on 12 real-world AUV trajectories collected in the Mediterranean Sea, comprising approximately 80.8 minutes of navigation data. Results show that SINNE reduces the mean position RMSE by 30.8%, 52.0%, and 64.8% for the EKF, UKF, and RI-IKF, respectively, compared with their fixed-covariance baselines. SINNE-RI-IKF achieves the best performance, reducing position RMSE by 42.4% relative to SINNE-UKF and by 57.2% relative to SINNE-EKF. These results highlight the benefit of combining adaptive process noise estimation with invariant filtering for underwater navigation.

Index Terms—Kalman filter, Sensor fusion, Adaptive process noise, State estimation, Lie group.

I. INTRODUCTION

Accurate positioning is a fundamental challenge in autonomous navigation, particularly in underwater environments where global navigation satellite system (GNSS) signals are unavailable. Autonomous underwater vehicles (AUVs) must therefore rely on onboard sensor fusion, commonly combining inertial measurements with Doppler velocity log (DVL) observations, to estimate position, velocity, and orientation.

Kalman-filter-based estimators, including the extended Kalman filter (EKF), unscented Kalman filter (UKF) [1], and right invariant Kalman filter (RI-IKF) [2], provide established frameworks for inertial/DVL sensor fusion [3], [4]. Their performance, however, depends strongly on the selection of the process noise covariance matrix $\mathbf{Q}$, which

is commonly assumed to be constant and manually tuned. In practical AUV operation, vehicle dynamics, maneuver intensity, sensor characteristics, and external disturbances vary over time. Consequently, a fixed $\mathbf{Q}$ may not adequately represent the uncertainty of the navigation model, resulting in inconsistent estimation and increased navigation error.

Classical approaches for addressing model uncertainty include innovation-based adaptive estimation [5]–[7], covariance-matching techniques [8], [9] and more recently neural adaptive approaches [10], [11]. These approaches adapt the process and/or measurement noise covariance matrices using statistical properties of the filter innovation. Although effective in many applications, their performance depends on the quality of the innovation statistics, the selected adaptation window, and several tuning parameters, which may limit their ability to respond to rapidly changing operating conditions.

In this work, we propose SINNE-KF, a general hybrid adaptive navigation framework that combines innovation-based covariance adaptation with data-driven neural estimation of time-varying inertial sensor noise. The neural network is trained on simulated inertial datasets generated from trajectories with varying motion profiles and sensor-noise levels, following the methodology introduced in [12]. The estimated inertial noise parameters are incorporated into the process noise covariance matrix and combined with information obtained from the filter innovations, allowing $\mathbf{Q}$ to adapt online to changing navigation conditions.

A central objective of this work is to examine how the same adaptive covariance-estimation mechanism interacts with different nonlinear filtering formulations. Accordingly, SINNE-KF is integrated into three navigation frameworks,

resulting in SINNE-EKF, SINNE-UKF, and SINNE-IKF. The three approaches are evaluated under identical sensor and adaptation conditions using real-world AUV trajectories collected in the Mediterranean Sea. This comparison allows us to determine whether the underlying filtering formulation influences the benefit obtained from adaptive process noise covariance estimation.

The main contributions of this article are:

- 1) A general hybrid adaptive process-noise estimation method combining innovation-based adaptation and neural inertial-noise estimation, applicable to the EKF, UKF, and RI-IKF.
- 2) A systematic real-world comparison of SINNE-EKF, SINNE-UKF, and SINNE-IKF on AUV trajectories, quantifying how different filtering formulations benefit from the same adaptive covariance-estimation mechanism.

Experimental evaluation on 12 A-KIT trajectories [13] shows that SINNE improves all three filtering frameworks, with the largest improvement obtained by the invariant formulation. In particular, SINNE-IKF achieves a mean position RMSE of 7.61 m compared with 13.21 m for SINNE-UKF, corresponding to a 42.4% reduction in position RMSE. The remainder of this paper is organized as follows. Section II introduces the EKF, UKF, and RI-IKF navigation formulations. Section III presents the proposed hybrid adaptive process noise estimation method, namely SINNE-KF. Section IV presents the experimental results, and Section V concludes the paper.

II. PROBLEM FORMULATION

Before we introduce the adaptive approach SINNE-KF that can be employed by EKF, UKF and RI-IKF. We will introduce how each estimator deals with non-linearity. In all of the following Kalman filter variants the inertial navigation equations are in ECEF frame.

$$\begin{aligned}\dot{\mathbf{p}}_{eb}^e &= \mathbf{v}_{eb}^e \\ \dot{\mathbf{v}}_{eb}^e &= \mathbf{C}_b^e \mathbf{f}_{ib}^b - 2(\boldsymbol{\omega}_{ie}^e)^\wedge \mathbf{v}_{eb}^e + \mathbf{g}_{ib}^e \\ \dot{\mathbf{C}}_b^e &= \mathbf{C}_b^e (\boldsymbol{\omega}_{ib}^b)^\wedge - (\boldsymbol{\omega}_{ie}^e)^\wedge \mathbf{C}_b^e\end{aligned}\quad (1)$$

where the $\mathbf{p}_{eb}^e$ and $\mathbf{v}_{eb}^e$ are the position and velocity respectively of the body frame relative to ECEF frame resolved in ECEF frame. $\boldsymbol{\omega}_{ib}^b$ and $\mathbf{f}_{ib}^b$ are the angular velocity and the unspecific force respectively of body with respect to ECI resolved in body. the matrix $\mathbf{C}_b^e$ is the rotation matrix from body frame to ECEF frame.

A. Extend Kalman filter

The INS error-state model is formulated in the Earth-centered, Earth-fixed (ECEF) frame following [4]. The error-state vector and the corresponding process-noise vector are defined as

$$\delta \mathbf{x} = \begin{bmatrix} \delta \phi^b \\ \delta \mathbf{v}_{eb}^e \\ \delta \mathbf{p}_{eb}^e \\ \delta \mathbf{b}_g \\ \delta \mathbf{b}_a \end{bmatrix} \in \mathbb{R}^{15} \quad \mathbf{w} = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{b_g} \\ \mathbf{n}_{b_a} \end{bmatrix} \in \mathbb{R}^{12} \quad (2)$$

where $\mathbf{w} \in \mathbb{R}^{12}$ consists of gyroscope noise, accelerometer noise, bias gyroscope noise and bias accelerometer noise. The error state dynamics comply with the following continuous differential equation

$$\delta \dot{\mathbf{x}} = \mathbf{F} \delta \mathbf{x} + \mathbf{G} \mathbf{w} \quad (3)$$

where the system matrix

$$\mathbf{F} = \begin{bmatrix} -\boldsymbol{\Omega}_{ie}^e & \mathbf{0} & \mathbf{0} & -\mathbf{C}_b^e & \mathbf{0} \\ -(\mathbf{C}_b^e \mathbf{f}_{ib}^b)^\wedge & -2\boldsymbol{\Omega}_{ie}^{e,\wedge} & \mathbf{g} - (\boldsymbol{\Omega}_{ie}^e)^2 & \mathbf{0} & -\mathbf{C}_b^e \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (4)$$

and the noise shaping matrix is

$$\mathbf{G} = \begin{bmatrix} -\mathbf{C}_b^e & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{C}_b^e & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I}_3 \end{bmatrix} \quad (5)$$

and the measurement model of the DVL is

$$\delta \mathbf{y} = \mathbf{H}_{\text{DVL}} \delta \mathbf{x} \quad (6)$$

where the measurement model matrix is

$$\mathbf{H}_{\text{DVL}} = [\mathbf{C}_e^b (\mathbf{v}_{eb}^e)^\wedge \quad \mathbf{C}_e^b \quad \mathbf{0}_{3 \times 9}] \quad (7)$$

B. Invariant Kalman filter

The right-invariant error-state model is formulated in ECEF frame following [2] [14] by convention

$$\boldsymbol{\eta}^R = \hat{\mathbf{X}} \mathbf{X}^{-1} = \exp((\boldsymbol{\xi}^R)^\wedge) \quad (8)$$

where

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!} \quad (9)$$

is the mapping from the Lie algebra $\mathfrak{se}_2(3)$ to the Lie group $SE_2(3)$ and the Lie discrepancy is

$$(\boldsymbol{\xi}^R)^\wedge = \begin{bmatrix} 0 & -\xi_3^R & \xi_2^R & \xi_4^R & \xi_7^R \\ \xi_3^R & 0 & -\xi_1^R & \xi_5^R & \xi_8^R \\ -\xi_2^R & \xi_1^R & 0 & \xi_6^R & \xi_9^R \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \in \mathfrak{se}_2(3) \quad (10)$$

and

$$\mathbf{X} = \begin{bmatrix} \mathbf{C}_b^e & \mathbf{v}_{eb}^e & \mathbf{p}_{eb}^e \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \in SE_2(3) \quad (11)$$

is the state encoded in Lie group. $\mathbf{X}$ is the GT state and $\hat{\mathbf{X}}$ is the estimated state.

Given that, the right invariant error is

$$\boldsymbol{\eta}^R = \begin{bmatrix} \hat{\mathbf{C}}_b^e \mathbf{C}_e^b & \hat{\mathbf{v}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{v}_{eb}^e & \hat{\mathbf{p}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{p}_{eb}^e \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \quad (12)$$

define the misalignment by

$$\hat{\mathbf{C}}_b^e \mathbf{C}_e^b = \exp(\phi^{e\wedge}) \approx I + \phi^{e\wedge} \quad (13)$$

and vectorize the right invariant error as

$$\delta \mathbf{x} = \begin{bmatrix} \phi^{e\wedge} \\ \hat{\mathbf{v}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{v}_{eb}^b \\ \hat{\mathbf{p}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{p}_{eb}^b \\ \delta \mathbf{b}_g \\ \delta \mathbf{b}_a \end{bmatrix} \in \mathbb{R}^{15} \quad \mathbf{w} = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{bg} \\ \mathbf{n}_{ba} \end{bmatrix} \in \mathbb{R}^{12} \quad (14)$$

where $\mathbf{w} \in \mathbb{R}^{12}$ is define as in (2).

The right invariant error state dynamics are

$$\delta \dot{\mathbf{x}} = \mathbf{F} \delta \mathbf{x} + \mathbf{G} \mathbf{w} \quad (15)$$

where the system matrix

$$\mathbf{F} = \begin{bmatrix} -\Omega_{ie}^e & \mathbf{0} & \mathbf{0} & \mathbf{C}_b^e & \mathbf{0} \\ -\mathbf{v}^\wedge \Omega_{ie}^e + \mathbf{g}^\wedge & -2\Omega_{ie}^e & \mathbf{0} & \mathbf{v}^\wedge \mathbf{C}_b^e & \mathbf{C}_b^e \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (16)$$

and the noise shaping matrix is

$$\mathbf{G} = \begin{bmatrix} \mathbf{C}_b^e & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{v}^\wedge \mathbf{C}_b^e & \mathbf{C}_b^e & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \quad (17)$$

where the measurement model of the RI-IKF is

$$\mathbf{H}_{\text{DVL}} = [\mathbf{0}_{3 \times 3} \quad -\mathbf{I}_3 \quad \mathbf{0}_{3 \times 9}] \quad (18)$$

For calculating the Kalman gain, posterior state covariance, posterior state read [14] or [12].

C. Unscented Kalman filter

Consider the initialized state vector $\mathbf{x}_0$ with a known mean and covariance:

$$\hat{\mathbf{x}}_0 = \mathbb{E}[\mathbf{x}_0], \quad (19)$$

$$\mathbf{P}_0 = \mathbb{E}[(\mathbf{x}_0 - \hat{\mathbf{x}}_0)(\mathbf{x}_0 - \hat{\mathbf{x}}_0)^\top]. \quad (20)$$

The weights for computing the mean (m upper case) and covariance (c upper case) are defined as:

$$w_0^{(m)} = \frac{\lambda}{n + \lambda}, \quad (21)$$

$$w_0^{(c)} = \frac{\lambda}{n + \lambda} + (1 + \alpha^2 + \beta), \quad (22)$$

$$w_i^{(m)} = w_i^{(c)} = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n, \quad (23)$$

where n is the dimension of the state vector and $\lambda = \alpha^2(n + \kappa) - n$. The parameters α , β , and κ control the spread and weighting of the sigma points and are addressed as hyperparameters. Typically, $\alpha = 10^{-3}$, $\kappa = 0$, and $\beta = 2$ for Gaussian distributions [15].

1) *Sigma Points*: The sigma points are constructed as

$$\chi_{0,k} = \hat{\mathbf{x}}_k, \quad i = 0 \quad (24)$$

$$\chi_{i,k} = \hat{\mathbf{x}}_k + \left[\sqrt{(n + \lambda) \mathbf{P}_k} \right]_i, \quad i = 1, \dots, n, \quad (25)$$

$$\chi_{i,k} = \hat{\mathbf{x}}_k - \left[\sqrt{(n + \lambda) \mathbf{P}_k} \right]_{i-n}, \quad i = n + 1, \dots, 2n, \quad (26)$$

where $\sqrt{\mathbf{P}_k}$ denotes the Cholesky decomposition of the covariance matrix. Each sigma point is propagated through the nonlinear model:

$$\chi_{i,k+1|k} = f(\chi_{i,k}), \quad i = 0, \dots, 2n. \quad (27)$$

The predicted state mean and covariance are then computed as

$$\hat{\mathbf{x}}_{k+1|k}^- = \sum_{i=0}^{2n} w_i^{(m)} \chi_{i,k+1|k} \quad (28)$$

$$\mathbf{P}_{k+1|k}^- = \sum_{i=0}^{2n} w_i^{(c)} (\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k}^-) (\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k}^-)^\top + \mathbf{Q}_{k+1}. \quad (29)$$

where $\mathbf{Q}_{k+1}$ is the covariance process noise matrix in time stamp $k+1$.

2) *Update step*: A new set of sigma points is generated from the predicted mean and covariance and propagated through the measurement function:

$$\mathcal{Z}_{i,k+1|k} = h(\chi_{i,k+1|k}), \quad i = 0, \dots, 2n. \quad (30)$$

And the measurement model of the DVL is:

$$\mathcal{Z}_{i,k+1|k} = h(\chi_{i,k+1|k}) \quad (31)$$

$$= \mathbf{C}_b^n (\mathbf{I} - \phi^{n\wedge}) [\mathbf{v}^n + \delta \hat{\mathbf{v}}_n^i] - \mathbf{C}_b^n \mathbf{v}^n \quad (32)$$

$$i = 0, \dots, 2n \quad (33)$$

The predicted measurement mean and covariance are computed as:

$$\hat{\mathbf{z}}_{k+1|k} = \sum_{i=0}^{2n} w_i^{(m)} \mathcal{Z}_{i,k+1|k} \quad (34)$$

$$\mathbf{S}_{k+1} = \sum_{i=0}^{2n} w_i^{(c)} (\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k}) (\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k})^\top + \mathbf{R}_{k+1}. \quad (35)$$

The cross-covariance matrix is define by:

$$\mathbf{P}_{xz,k+1} = \sum_{i=0}^{2n} w_i^{(c)} (\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k}^-) (\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k})^\top \quad (36)$$

The Kalman gain is computed by using (35) and (36):

$$\mathbf{K}_{k+1} = \mathbf{P}_{xz,k+1} \mathbf{S}_{k+1}^{-1} \quad (37)$$

Finally, the state mean and covariance are updated using the measurement $\mathbf{z}_{k+1}$ and the Kalman gain (37):

$$\hat{\mathbf{x}}_{k+1}^+ = \hat{\mathbf{x}}_{k+1|k}^- + \mathbf{K}_{k+1}(\mathbf{z}_{k+1} - \hat{\mathbf{z}}_{k+1|k}) \quad (38)$$

$$\mathbf{P}_{k+1}^+ = \mathbf{P}_{k+1|k}^- - \mathbf{K}_{k+1}\mathbf{S}_{k+1}\mathbf{K}_{k+1}^T \quad (39)$$

III. PROPOSED APPROACH

In order to address the limitation of the fixed process noise covariance. We propose a deep learning neural network that is trained on simulation data to dynamically estimate the process noise covariance matrix. This approach enable the filter to adapt to changing condition, improving robustness and maintaining accurate position over time. Since different navigation filtering frameworks handle nonlinear dynamics differently. Applying the same adaptive process noise method may lead to different levels of improvement across the various filtering frameworks. We therefore aim to determine which framework benefits the most from an accurately estimated process noise covariance matrix.

We propose the sim2real inertial neural adaptive process covariance learning network for Kalman filtering (SINNE-KF). In which we generate simulated inertial datasets based on trajectories with varying motion profiles and sensor noise levels, consistent with the methodology introduced in [12]. In this framework, a light and efficient neural network is trained to estimate the standard deviation of the accelerometer and gyroscope measurements. During testing, the trained model is evaluated on real data collected in trials conducted in the Mediterranean sea, from sensors mounted on an AUV. The estimated noise parameters are then used within one of the filtering framework EKF, UKF or RI-IKF to improve state estimation accuracy under real-world conditions. The overall pipeline of the proposed method is illustrated in Fig. 1.

The adaptive process covariance of SINNE-KF is consists of two parts. The first part is the data-driven sim2real neural network that utilize past inertial window to learn the process noise covariance and all the Kalman filter SINNE-KF variants that is EKF-SINNE, UKF-SINNE and IKF-SINNE, uses the same neural network with the same weights. The second part is the model-based approach that exploits past innovation window in order to extract process noise covariance matrix namely the adaptive innovation based approach. For the UKF-SINNE and EKF-SINNE our approach follows the approach of [5] but for the adaptive innovation based IKF-SINNE we have imployed the approach from [12].

A. Data-driven neural inertial learning approach

The neural-learning-based approach in [12] maps a window of recent inertial measurements to the gyroscope and accelerometer process-noise parameters:

$$\mathbf{q} = f_{\theta}(\mathcal{W}) = \begin{bmatrix} \mathbf{q}_g \\ \mathbf{q}_a \end{bmatrix} \in \mathbb{R}^6, \quad (40)$$

where f_{θ} denotes the neural network, $\mathcal{W}$ is a window of recent inertial measurements, and $\mathbf{q}_g, \mathbf{q}_a \in \mathbb{R}^3$ contain the

predicted process-noise variances of the gyroscope and accelerometer, respectively. The continuous-time process-noise covariance matrix is then defined as

$$\mathbf{Q}_{c,\text{neural}} = \text{diag}(\mathbf{q}_g, \mathbf{q}_a, \mathbf{q}_{bg}, \mathbf{q}_{ba}) \in \mathbb{R}^{12 \times 12}, \quad (41)$$

where $\mathbf{q}_{bg}, \mathbf{q}_{ba} \in \mathbb{R}^3$ represent the fixed gyroscope-bias and accelerometer-bias process-noise variances, respectively.

B. Adaptive innovation model-based approach EKF/UKF

The sensor fusion algorithm EKF and UKF has Euclidean innovation which is the difference between the DVL measurement and the estimated velocity in body frame which is

$$\nu_{\text{innov}} = v_{eb}^b - \hat{v}_{eb}^b \quad (42)$$

where v_{eb}^b is the velocity in body frame as measured by the DVL and $\hat{v}_{eb}^b$ is the estimated velocity in body as given by the EKF or UKF.

The adaptive innovation-based process noise in steady state is given by [5]

$$\mathbf{Q}_{\text{innov}} = \frac{1}{N} \sum_{n=0}^N \mathbf{K}_n \nu_n \nu_n^T \mathbf{K}_n^T \quad (43)$$

where $\mathbf{K}_n$ is the Kalman gain at step n given by the EKF or the UKF and ν_n is the innovation as given by (42) at step n

C. Adaptive innovation model-based approach RI-IKF

While the EKF and UKF has difference between the prior and the posterior is given by

$$\mathbf{x} - \hat{\mathbf{x}} = \mathbf{K}\nu \quad (44)$$

where $\mathbf{x} \in \mathbb{R}^{15}$ is the posterior state, $\hat{\mathbf{x}} \in \mathbb{R}^{15}$ is the prior state and $\mathbf{K}$ is the Kalman gain.

The RI-IKF difference between the prior and posterior state is given by (8) which is

$$(\xi^R)^\wedge = \log(\hat{\mathbf{X}}\mathbf{X}^{-1}) \quad (45)$$

where the function

$$\log(A) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (A - I)^k \quad (46)$$

is a nonlinear mapping from the Lie group $SE_2(3)$ into the Lie algebra $\mathfrak{se}_2(3)$ and $\mathbf{X}$ is given from (11)

As shown in [12] the adaptive innovation-based process noise covariance that is derived from Lie innovation is

$$\mathbf{Q}_{\text{innov}} = \frac{1}{N} \sum_{n=1}^N \log(\hat{\mathbf{X}}\mathbf{X}^{-1})^\vee \log(\hat{\mathbf{X}}\mathbf{X}^{-1})^{\vee,T} \quad (47)$$

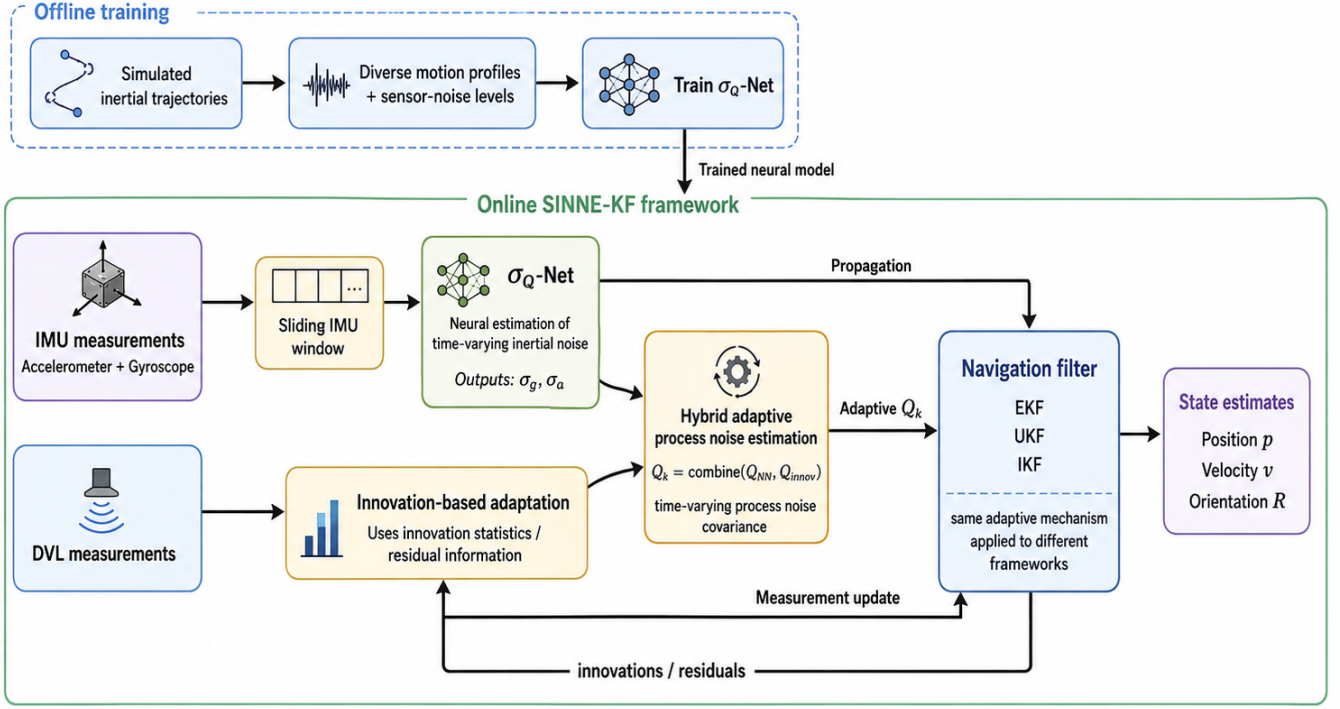


Fig. 1. Overview of the proposed SINNE-KF framework. A neural network estimates time-varying inertial sensor noise from a sliding IMU window, while an innovation-based adaptive mechanism exploits filter residual statistics. The two sources are combined to construct an adaptive process noise covariance matrix $\mathbf{Q}_k$, which is incorporated into the EKF, UKF, and IKF navigation frameworks for IMU/DVL sensor fusion.

where $\mathbf{X}$ is the posterior state and $\hat{\mathbf{X}}$ is the prior state and the mapping $(\cdot)^V$ is given by

$$\begin{bmatrix} 0 & -\xi_3^R & \xi_2^R & \xi_4^R & \xi_7^R \\ \xi_3^R & 0 & -\xi_1^R & \xi_5^R & \xi_8^R \\ -\xi_2^R & \xi_1^R & 0 & \xi_6^R & \xi_9^R \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}^V = \begin{bmatrix} \xi_1^R \\ \xi_2^R \\ \xi_3^R \\ \xi_4^R \\ \xi_5^R \\ \xi_6^R \\ \xi_7^R \\ \xi_8^R \\ \xi_9^R \end{bmatrix} \quad (48)$$

D. The SINNE approach

Ultimately, The discrete-time process-noise covariance used by all Kalman filter variant that is EKF-SINNE, UKF-SINNE and IKF-SINNE, is obtained by averaging the neural-network-based covariance and the adaptive innovation-based covariance:

$$\mathbf{Q}_d = \frac{1}{2} (\mathbf{G}\mathbf{Q}_{c,\text{neural}}\mathbf{G}^T + \mathbf{Q}_{d,\text{innovation}}) \quad (49)$$

where the neural process noise covariance $\mathbf{Q}_{c,\text{neural}}$ is obtained by the same neural network with the same weights in all variants EKF-SINNE, UKF-SINNE and IKF-SINNE. The innovation based covariance in EKF-SINNE and UKF-SINNE given by (43) and the innovation based covariance $\mathbf{Q}_{d,\text{innovation}}$ in IKF-SINNE is given by (47).

IV. ANALYSIS AND RESULTS

The dataset used for the evaluation of our results was acquired by the Snapir AUV during maneuvers in the Mediterranean Sea [13], with a total mission duration of 4,800 seconds. To ensure a consistent and controlled evaluation across all filtering frameworks, additional white noise was injected into the inertial and DVL measurements. The accelerometer measurements were corrupted with white noise having a standard deviation of 4×10^{-3} , m/s², the gyroscope measurements with a standard deviation of 1×10^{-4} , rad/s, and the DVL measurements with a standard deviation of 5×10^{-3} , m/s. The same noise realization and noise levels were applied consistently to all evaluated filtering methods to ensure a fair comparison.

We evaluate the filtering frameworks EKF, UKF and RI-IKF with and without the proposed SINNE-KF against the post-processed reference solution. Table I shows the performance of each navigation frameworks in terms of root mean square error (RMSE) for the position (PRMSE), velocity (VRMSE) and heading angle (YRMSE).

The results demonstrate a clear and consistent improvement when incorporating the SINNE approach across all filtering methods. In terms of position accuracy (PRMSE), every baseline filter benefits significantly from SINNE, with reductions of approximately 31% for EKF, 52% for UKF, and 65% for RI-IKF, with the RI-IKF showing the most substantial improvement. A similar trend is observed in the

TABLE I
COMPARISON OF FILTERING METHODS FOR THE SNAPIR AUV,
AVERAGED OVER 12 TRAJECTORIES

Method	PRMSE [m]	VRMSE [m/s]	YRMSE [deg]
EKF	25.71	0.22	3.85
EKF SINNE	17.80	0.18	3.44
UKF	27.51	0.25	4.01
UKF SINNE	13.21	0.18	3.52
RI-IKF	21.64	0.24	3.42
RI-IKF SINNE	7.61	0.17	3.32

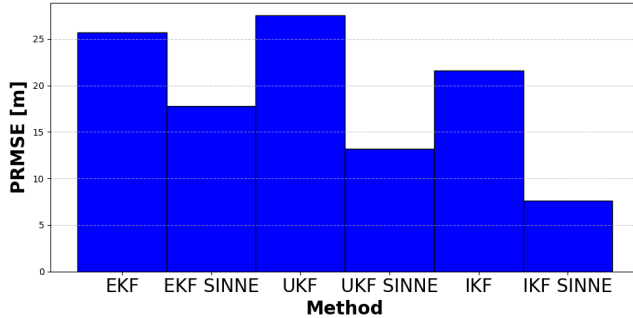


Fig. 2. PRMSE of each method: EKF, SINNE-EKF, UKF, SINNE-UKF, RI-IKF, SINNE-IKF

velocity estimates (VRMSE), where SINNE variants reduce the error by about 18–28%, with IKF-SINNE achieving the best performance at 0.17 m/s. For heading estimation (YRMSE), the improvements are more moderate, on the order of 5–11%, indicating enhanced stability while maintaining accurate orientation estimates. Overall, the UKF-SINNE and especially the RI-IKF SINNE significantly outperform their standard counterparts, highlighting the effectiveness of the adaptive approach in reducing estimation errors and improving robustness across all evaluated states. A histogram of PRMSE for each of the methods is shown in Figure 2.

V. CONCLUSION

This paper presented SINNE-KF, a hybrid adaptive navigation framework that combines innovation-based covariance adaptation with neural estimation of time-varying inertial sensor noise. The proposed method was incorporated into three nonlinear filtering frameworks, resulting in SINNE-EKF, SINNE-UKF, and SINNE-RI-IKF. This formulation enabled a direct comparison of how different Kalman filtering approaches benefit from the same adaptive process noise covariance estimation mechanism.

The proposed approaches were evaluated on 12 real-world AUV trajectories collected in the Mediterranean Sea, with a total navigation duration of 4,800 seconds. To provide a controlled and consistent comparison, additional white noise was applied to the inertial and DVL measurements, and identical noise conditions were used for all evaluated filtering frameworks. The experimental results demonstrated that adaptive process noise estimation improves the position accuracy of all three filtering approaches. Compared with

their corresponding fixed-covariance baselines, SINNE reduced the mean position RMSE by approximately 30.8%, 52.0%, and 64.8% for the EKF, UKF, and IKF, respectively. Among the evaluated methods, SINNE-IKF achieved the best overall navigation performance. In particular, its mean position RMSE was reduced by 42.4% relative to SINNE-UKF and by 57.2% relative to SINNE-EKF. The results therefore indicate that the benefit obtained from adaptive process noise estimation depends strongly on the underlying filtering formulation. In particular, the invariant error representation appears to provide a more suitable framework for exploiting time-varying covariance information in nonlinear inertial/DVL navigation.

These findings demonstrate that combining model-based innovation information with data-driven inertial noise estimation can substantially improve underwater navigation performance without restricting the adaptation mechanism to a particular Kalman filter formulation. Future work will investigate joint adaptation of the process and measurement noise covariance matrices, evaluate the proposed framework under additional vehicle dynamics and environmental conditions, and study its real-time implementation and computational requirements for onboard AUV navigation.

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